Quantum Mechanics

Lecture 3: Operators, Measurement, and Wave Functions

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Transcript-derived companion notes curated by LazyingArt LLC
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Chapter 1

Operators, Measurement, and Wave Functions

Quantum mechanics is linear algebra. A state is a vector, while a measurable quantity is represented by a special kind of linear operator. The purpose of this lecture is to connect that abstract statement to calculations. We shall first turn operators into matrices, then identify Hermitian operators as observables, state the measurement postulates, and finally apply the entire structure to a particle moving on a line.

1.1 Operators, Matrix Elements, and Bases

A linear operator maps one ket into another:

$$\hat{K}\left|A\right\rangle = \left|C\right\rangle . \tag{1.1}$$

Once a bra is supplied, the result becomes a complex number,

$$\left\langle B\right|\hat{K}\left|A\right\rangle . \tag{1.2}$$

The order is important. The operator first acts on $|A\rangle$; the resulting ket is then paired with $\langle B|$.

Choose an orthonormal basis $\{|n\rangle\}_{n=1}^{d}$. An ordinary three-dimensional sketch helps us remember what a basis does, although a quantum state space is generally complex and may have far more than three dimensions.

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There are many possible bases. A vector drawn in one basis can always be expanded in another complete basis; only its components change.

Question from the audience. Can a vector associated with one drawn basis be represented in the other, and do the labels m and n refer to different bases?[a]

Response. Yes, the same vector can be expanded in either complete basis. In the matrix element below, however, both m and n are labels drawn from one chosen basis. They identify the row and column of the matrix representing $\hat{K}$ in that basis.

[a] Lecture timestamp: 00:05:24–00:07:40.

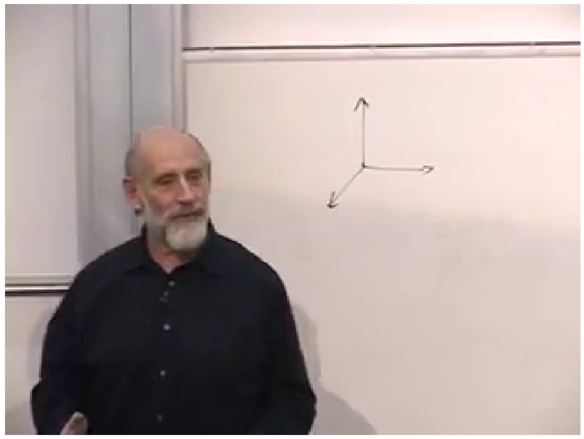

Figure 1.1: An ordinary vector sketch standing in for an orthonormal basis.

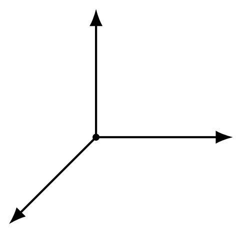

Figure 1.2: A clean reconstruction of the three-direction basis sketch.

The basis matrix elements are

$$K_{mn} := \langle m| \hat{K} |n\rangle . \tag{1.3}$$

There are d^2 of them in a d-dimensional space, and together they determine the operator once the basis has been fixed.

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1.2 From Kets to Matrix Multiplication

Every ket can be expanded in the chosen basis:

$$|A\rangle = \sum_n A_n |n\rangle , \qquad A_n = \langle n \,|\, A\rangle . \tag{1.4}$$

Substitution gives

$$|A\rangle = \sum_n |n\rangle \langle n| \, |A\rangle . \tag{1.5}$$

Since this holds for every $|A\rangle$, the operator between the summation sign and the final ket must be the identity:

$$\sum_n |n\rangle \langle n| = \mathbb{I}. \tag{1.6}$$

This is the completeness relation. The adjacent ket and bra form a projector onto one basis direction; summing all such projectors does nothing to the vector.

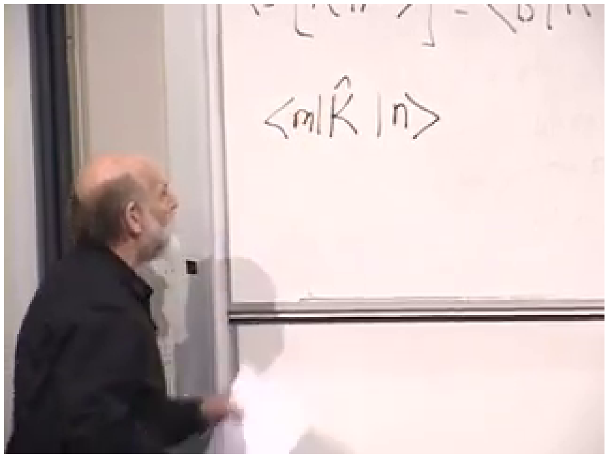

Figure 1.3: The matrix element $\langle m|\hat{K}|n\rangle$ on the board.

We can now calculate the components of $\hat{K}|A\rangle$:

$$\langle n|\hat{K}|A\rangle = \sum_m \langle n|\hat{K}|m\rangle \langle m \mid A\rangle \tag{1.7}$$

$$= \sum_m K_{nm} A_m. \tag{1.8}$$

Thus the abstract operation $\hat{K}|A\rangle$ is ordinary matrix multiplication when written in components.

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The same correspondence holds for products of operators. In $\hat{K}\hat{L}$, $\hat{L}$ acts first and $\hat{K}$ acts second. Inserting Eq. (1.6) between them gives

$$(KL)_{nm} = \langle n|\hat{K}\hat{L}|m\rangle \tag{1.9}$$

$$= \sum_r \langle n|\hat{K}|r\rangle \langle r|\hat{L}|m\rangle \tag{1.10}$$

$$= \sum_r K_{nr} L_{rm}. \tag{1.11}$$

This is precisely the rule for multiplying matrices. It also makes clear why operator order matters: in general $\hat{K}\hat{L} \neq \hat{L}\hat{K}$.

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1.3 Hermitian Operators and Their Eigenvectors

The operators associated with observables must have real measurement values. The relevant class is the Hermitian operators. Here the letter H merely stands for Hermitian; it is not yet the Hamiltonian.

Definition 1.1. An operator $\hat{H}$ is Hermitian when

$$\langle B|\hat{H}|A\rangle = \Big(\langle A|\hat{H}|B\rangle\Big)^* \tag{1.12}$$

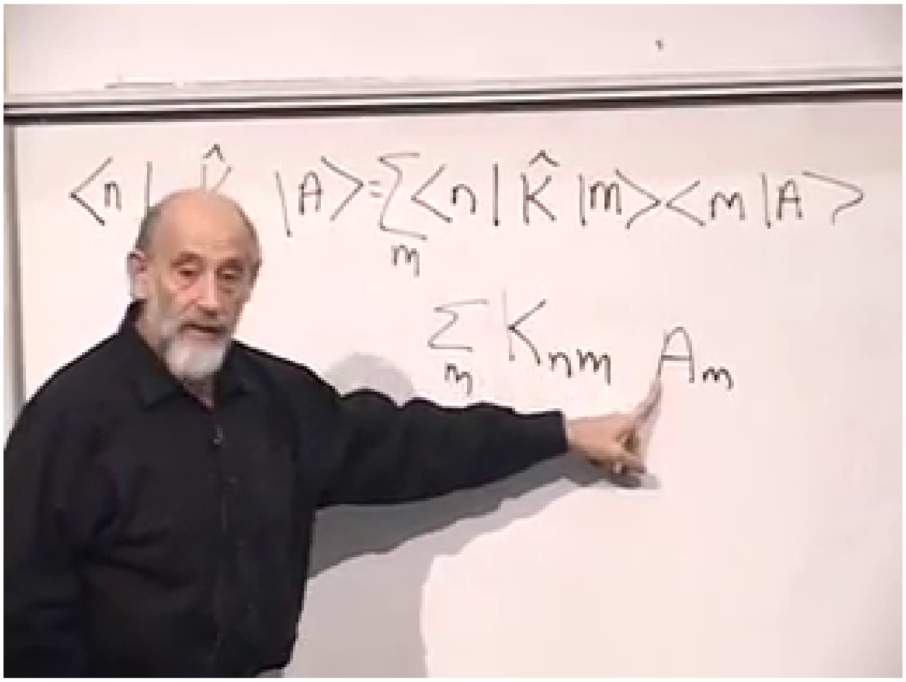

Figure 1.4: The passage from operator action to the component sum $\sum_m K_{nm}A_m$.

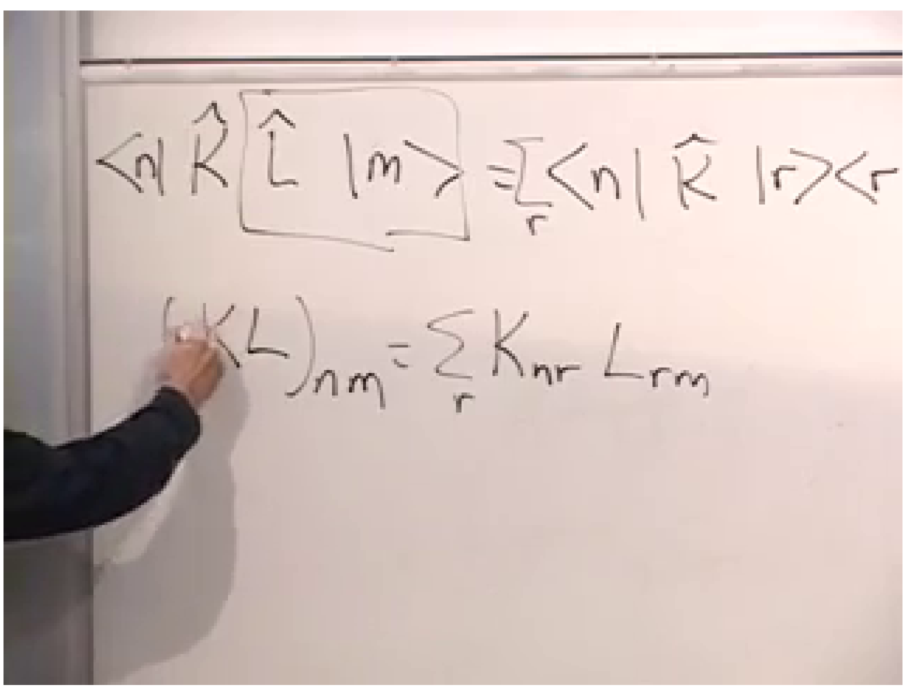

Figure 1.5: Insertion of a complete basis between two operators.

for vectors in its domain. In finite dimensions this is equivalent to saying that $\langle A|\,\hat{H}\,|A\rangle$ is real for every $|A\rangle$.

Most vectors change direction under a general operator. An eigenvector is a special direction that does not:

$$\hat{H}\,|\lambda\rangle = \lambda\,|\lambda\rangle\,. \tag{1.13}$$

Only its length and phase may change, through the scalar eigenvalue λ.

Question from the audience. Does every operation have eigenvectors? [a]

Response. Over a finite-dimensional complex vector space, every linear operator has at least one eigenvector, but a general operator need not have enough eigenvectors to form a basis, much less an orthonormal basis. A real rotation can also lack real eigenvectors. The special result needed here is stronger: a Hermitian operator has a complete orthonormal eigenbasis. In infinite-dimensional spaces, continuous spectra require generalized eigenvectors, as the position example will show.

[a] Lecture timestamp: 00:29:53–00:30:18.

The finite-dimensional spectral theorem supplies three facts:

1. every eigenvalue of a Hermitian operator is real;
2. eigenvectors with distinct eigenvalues are orthogonal;
3. the space has a complete orthonormal basis of eigenvectors.

The first two statements follow directly from Hermiticity. If $\hat{H}\,|\lambda\rangle = \lambda\,|\lambda\rangle$, then

$$\langle\lambda|\,\hat{H}\,|\lambda\rangle = \lambda\,\langle\lambda\,|\,\lambda\rangle\,. \tag{1.14}$$

The left side is real and the norm is positive, so λ is real. For two eigenvectors,

$$\langle\lambda_1|\,\hat{H}\,|\lambda_2\rangle = \lambda_2\,\langle\lambda_1\,|\,\lambda_2\rangle\,, \tag{1.15}$$

$$\langle\lambda_1|\,\hat{H}\,|\lambda_2\rangle = \lambda_1\,\langle\lambda_1\,|\,\lambda_2\rangle\,. \tag{1.16}$$

Consequently,

$$(\lambda_2 - \lambda_1)\,\langle\lambda_1\,|\,\lambda_2\rangle = 0. \tag{1.17}$$

Distinct eigenvalues therefore give orthogonal eigenvectors. If an eigenvalue is degenerate, an orthonormal basis can be chosen within its eigenspace. Finally, every eigenvector can be rescaled to unit norm.

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Question from the audience. Is a rotation Hermitian? [a]

Response. A finite rotation is generally unitary, or orthogonal in a real vector space, rather than Hermitian. Its infinitesimal generator is anti-Hermitian; equivalently, the generator can be written as $-i$ times a Hermitian operator. In a real basis, symmetric matrices are the analogues of Hermitian matrices, while infinitesimal rotation generators are antisymmetric.

[a] Lecture timestamp: 00:35:35–00:36:55.

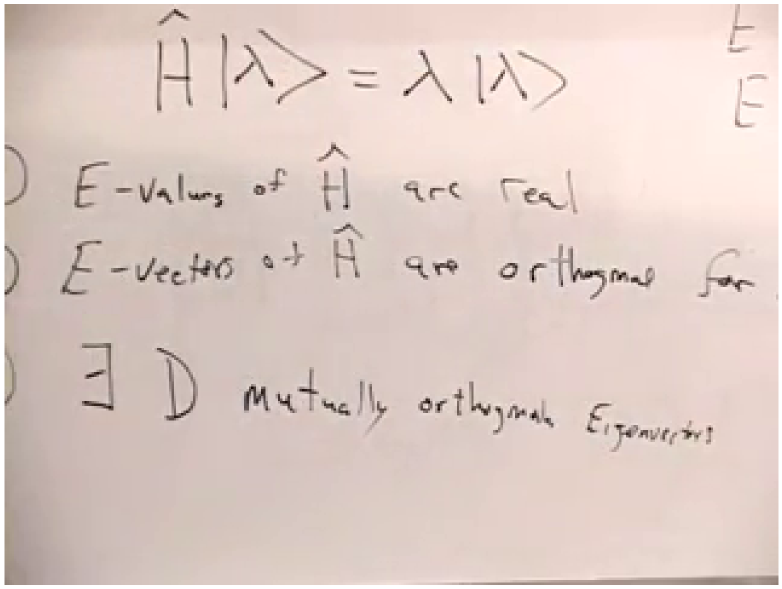

Figure 1.6: The eigenvalue equation and the three spectral facts for a Hermitian operator.

Question from the audience. In which vector space do the eigenvectors form a basis? [a]

Response. They form a basis of the same complex vector space on which $\hat{H}$ acts. If we choose that eigenbasis for our coordinates, the matrix of $\hat{H}$ is diagonal, with its eigenvalues along the diagonal.

[a] Lecture timestamp: 00:37:35–00:40:05.

1.4 Five Measurement Postulates

The basic rules can now be stated in the same language.

1. A physical state is represented by a ket in a complex vector space. A coin has a two-dimensional state space, a die has a six-dimensional one, and a particle on a line requires an infinite-dimensional space.
2. Every observable is represented by a Hermitian operator.
3. The possible outcomes of measuring an observable are the eigenvalues of its operator.
4. An eigenvector $|\lambda_n\rangle$ is a state in which that observable has the definite value λ_n.
5. If the normalized state is $|A\rangle$, the probability of obtaining λ_n is

$$P(\lambda_n \mid A) = |\langle \lambda_n \mid A\rangle|^2 . \tag{1.18}$$

Lecture frame: 00:54:50.

If the eigenvectors form an orthonormal basis, normalization guarantees that the probabilities sum to one:

$$\sum_n P(\lambda_n \mid A) = \sum_n |\langle \lambda_n \mid A\rangle|^2 = \langle A \mid A\rangle = 1. \tag{1.19}$$

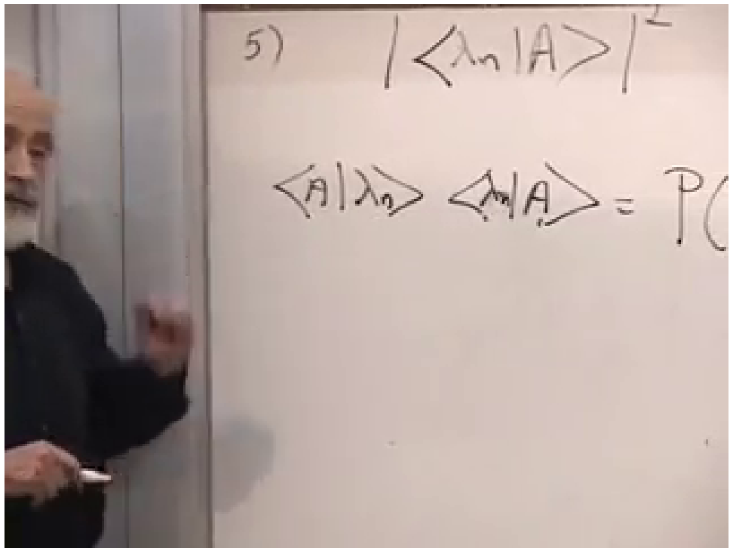

Figure 1.7: The fifth postulate written as a squared projection onto an eigenvector.

Question from the audience. Is a state itself an observable? [a]

Response. No. Position, momentum, spin components, and fields are examples of observables. A state is not a number read from an apparatus; it encodes the probabilities for the possible readings. A single measurement therefore does not reveal the complete state.

[a] Lecture timestamp: 00:42:05–00:42:50.

Question from the audience. If eigenvalues are real, what about experimental error bars?[a]

Response. Real eigenvalues do not eliminate experimental uncertainty. An apparatus may report a real interval or a real value with an error bar. The point is that a measurement outcome is not an imaginary number.

[a] Lecture timestamp: 00:45:33–00:46:28.

Question from the audience. Do we observe eigenvectors, and what does it mean for an observable to be definite?[a]

Response. The apparatus records an eigenvalue, not an eigenvector. An observable is definite when repeated ideal measurements on identically prepared systems give the same value with probability one. For example, a spin state can be definite along one axis and probabilistic along another, because it is an eigenstate of one spin operator but not of the other.

[a] Lecture timestamp: 00:47:41–00:50:29.

1.5 A Particle on a Line

We now leave finite-dimensional examples and consider a particle moving on the real line. Its states are represented by complex functions $\psi(x)$. Functions form a vector space, and the natural inner product is

$$\langle \phi \,|\, \psi \rangle = \int_{-\infty}^{\infty} \phi^*(x)\psi(x)\,dx. \tag{1.20}$$

We shall initially work with square-integrable, sufficiently well-behaved functions and impose whatever boundary conditions are needed by the operators.

Question from the audience. Is the inner product of two wave functions always real? [a]

Response. No. The inner product $\langle \phi \,|\, \psi \rangle$ can be complex. What must be real and nonnegative is a vector's inner product with itself:

$$\langle \psi \,|\, \psi \rangle = \int_{-\infty}^{\infty} |\psi(x)|^2 \, dx.$$

[a]Lecture timestamp: 00:58:13–00:59:07.

1.5.1 The position operator

The position operator multiplies a wave function by its argument:

$$(\hat{x}\psi)(x) = x\psi(x). \tag{1.21}$$

It is Hermitian on its proper domain because

$$\langle \psi | \, \hat{x} \, | \psi \rangle = \int_{-\infty}^{\infty} x |\psi(x)|^2 \, dx \tag{1.22}$$

is real.

The position eigenvalue equation is

$$\hat{x} \, |\lambda\rangle = \lambda \, |\lambda\rangle \, , \tag{1.23}$$

or, in the x-representation,

$$(x - \lambda)\psi_\lambda(x) = 0. \tag{1.24}$$

For every $x \neq \lambda$, the factor $x - \lambda$ is nonzero, so the eigenfunction must vanish. Its support is concentrated at the single point $x = \lambda$.

Question from the audience. Have we assumed that the norm integral converges? [a]

Response. Yes. Physical state vectors are normalized, so their wavefunctions are square-integrable. For the elementary manipulations here we also assume sufficiently tame behavior at large distance. Square integrability by itself does not force pointwise decay in every mathematical sense, but the chosen operator domains will supply the boundary behavior needed below.

[a]Lecture timestamp: 01:05:02–01:05:42.

No ordinary square-integrable function can be nonzero at only one point and have unit norm. The position eigenfunction is therefore a generalized function:

$$\langle x \,|\, \lambda \rangle = \delta(x - \lambda). \tag{1.25}$$

The lecture pictures $\delta(x - \lambda)$ as a rectangle of width ϵ and height $1/\epsilon$, then lets ϵ shrink. This choice gives unit area,

$$\int_{-\infty}^{\infty} \delta(x - \lambda) \, dx = 1, \tag{1.26}$$

but it does not give unit Hilbert-space norm: the integral of the square of the rectangle grows like $1/\epsilon$. The delta function is not an ordinary normalized state.

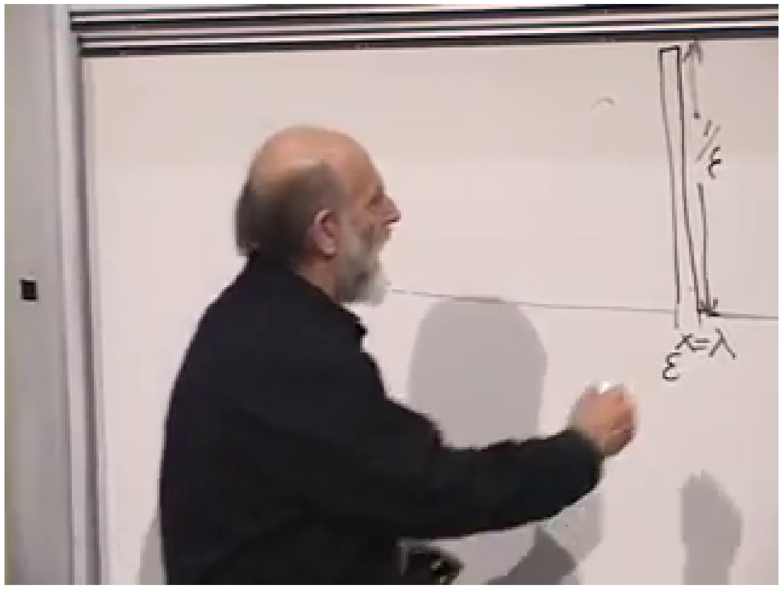

Figure 1.8: A narrow unit-area approximation to $\delta(x-\lambda)$.

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Question from the audience. Are these position eigenfunctions orthonormal? [a]

Response. They are orthogonal in the generalized continuous sense, but not normalizable one by one as ordinary vectors. The precise replacement for Kronecker-delta normalization is

$$\langle x \mid x' \rangle = \delta(x - x').$$

Position eigenkets are delta-normalized generalized eigenvectors.

[a] Lecture timestamp: 01:09:13–01:11:22.

There is one such generalized eigenvector for every real position, so the spectrum of $\hat{x}$ is continuous. Completeness now takes the form

$$\int_{-\infty}^{\infty} |x\rangle \langle x| \; dx = \mathbb{I}. \tag{1.27}$$

1.5.2 What the wave function means

The delta function samples an ordinary function:

$$\int_{-\infty}^{\infty} \delta(x-\lambda) f(x)\, dx = f(\lambda). \tag{1.28}$$

Applying this to a state gives

$$\langle \lambda \mid \psi \rangle = \int_{-\infty}^{\infty} \delta(x-\lambda)\psi(x)\, dx = \psi(\lambda). \tag{1.29}$$

Relabeling the eigenvalue by x,

$$\psi(x) = \langle x \mid \psi \rangle . \tag{1.30}$$

The wave function is therefore the component of the abstract state ket in the position basis.

The fifth postulate then says that

$$\rho(x) = |\psi(x)|^2 \tag{1.31}$$

is the position probability density. A probability belongs to an interval, not to one isolated point:

$$P(a \leq x \leq b) = \int_a^b |\psi(x)|^2 \, dx. \tag{1.32}$$

Question from the audience. Has ψ changed from an arbitrary function into one special function?[a]

Response. No. Any normalized state wavefunction in the allowed domain supplies a probability distribution through $|\psi(x)|^2$. The wavefunction itself may be complex and may carry signs and phases. Its absolute square is real and nonnegative. Those phases are essential because position is not the only observable we may measure; replacing ψ by the probability density alone would discard information needed for other measurements.

[a] Lecture timestamp: 01:18:43–01:19:28.

1.6 Differentiation and the Momentum Operator

Differentiation is another linear operation on functions:

$$\frac{d}{dx}(\alpha\psi + \beta\phi) = \alpha\frac{d\psi}{dx} + \beta\frac{d\phi}{dx}. \tag{1.33}$$

The derivative itself is not Hermitian. With boundary terms chosen to vanish, it is anti-Hermitian:

$$\left(\frac{d}{dx}\right)^\dagger = -\frac{d}{dx}. \tag{1.34}$$

Multiplication by $-i$ converts it into a Hermitian operator,

$$\hat{K} = -i\frac{d}{dx}. \tag{1.35}$$

The key calculation is integration by parts. For functions in a common domain with $\phi^*(x)\psi(x) \to 0$ at both ends,

$$\begin{aligned}
\langle\phi|\,\hat{K}\,|\psi\rangle &= -i\int_{-\infty}^{\infty} \phi^*(x)\frac{d\psi}{dx}\,dx && (1.36)\\
&= -i\,[\phi^*(x)\psi(x)]_{-\infty}^{\infty} + i\int_{-\infty}^{\infty} \frac{d\phi^*}{dx}\psi(x)\,dx && (1.37)\\
&= \left\langle \hat{K}\phi \,\middle|\, \psi \right\rangle. && (1.38)
\end{aligned}$$

Thus $\hat{K}$ is Hermitian on that domain. The boundary conditions are part of the operator; without them, the formal differential expression alone is not enough to establish Hermiticity.

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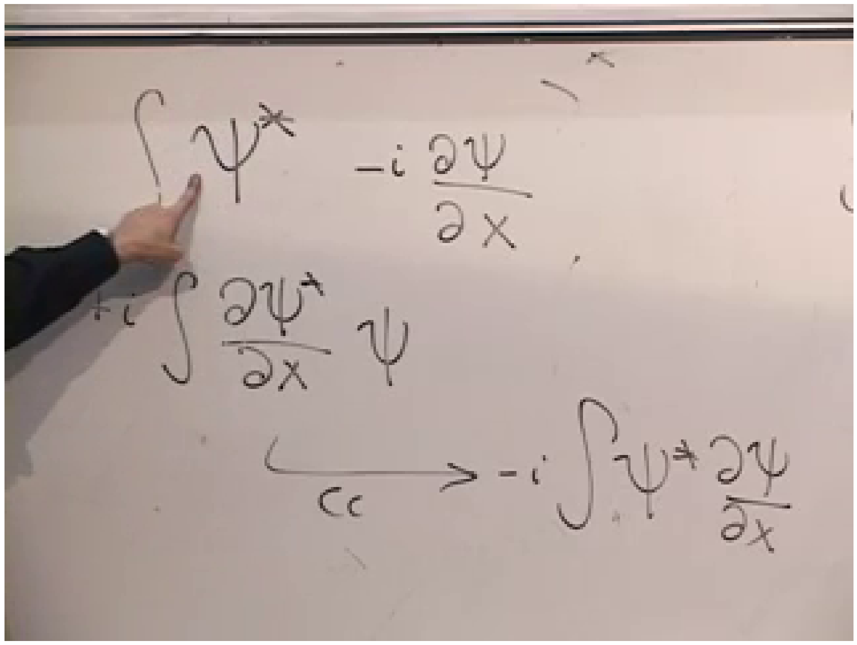

Figure 1.9: The integration-by-parts step that moves the derivative and changes its sign.

1.6.1 Plane-wave eigenfunctions

The eigenvalue equation

$$-i\frac{d\psi_k}{dx} = k\psi_k \tag{1.39}$$

has the solution

$$\psi_k(x) \propto e^{ikx} = \cos(kx) + i\sin(kx). \tag{1.40}$$

Indeed,

$$-i\frac{d}{dx}e^{ikx} = -i(ik)e^{ikx} = ke^{ikx}. \tag{1.41}$$

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A plane wave is the opposite of a position eigenstate. Its magnitude is constant:

$$|e^{ikx}|^2 = 1. \tag{1.42}$$

Formally, it gives no preferred position at all. It is also not square-integrable on the whole line, so, like the delta-localized position eigenket, it is a generalized eigenstate rather than a normalizable physical state by itself.

Question from the audience. Did the Hermiticity argument not assume decay at infinity, while the plane waves fail to decay?[a]

Response. Yes. The tension is real and should not be hidden. Plane waves belong to the generalized continuous spectrum. One may approximate them by long, normalizable wave packets, work in a large periodic box and then take its size to infinity, or use delta normalization. The formal plane waves remain useful because they expose the spectral structure exactly.

[a] Lecture timestamp: 01:38:41–01:39:47.

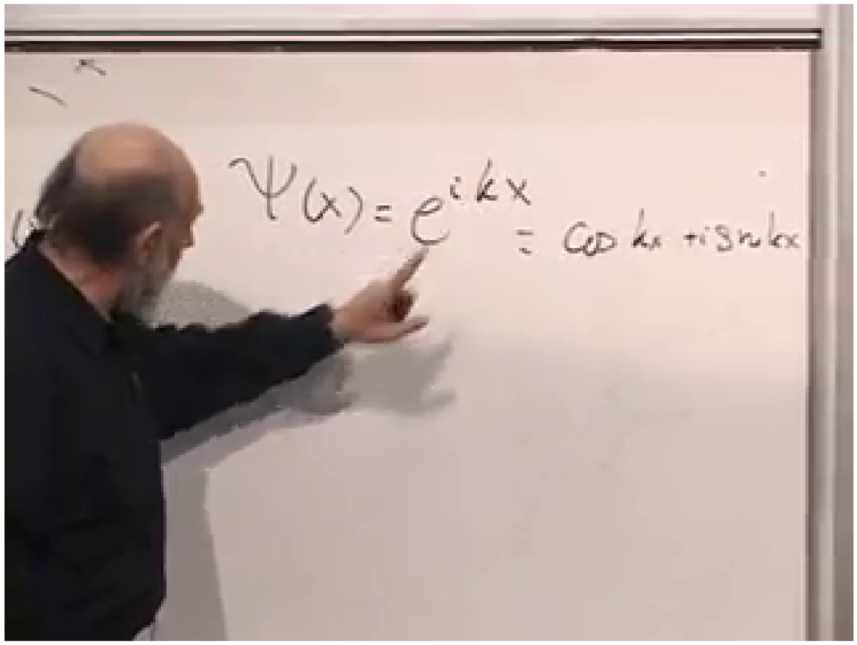

Figure 1.10: The plane-wave eigenfunction of $\hat{K} = -i\,d/dx$.

1.7 Why Wave Number Measures Momentum

One full oscillation of e^{ikx} occurs when its phase changes by 2π. If L denotes the wavelength, then

$$kL = 2\pi, \qquad L = \frac{2\pi}{|k|}. \tag{1.43}$$

The sign of k distinguishes the two propagation directions, while its magnitude fixes the wavelength. Einstein's photon relation and de Broglie's extension to material particles suggest

$$p = \frac{h}{L}. \tag{1.44}$$

Using $L = 2\pi/|k|$ for the magnitude and retaining the sign for direction,

$$p = \hbar k, \qquad \hat{p} = \hbar\hat{K} = -i\hbar\frac{d}{dx}. \tag{1.45}$$

Lecture frame: 01:43:50.

At this stage, Eq. (1.45) is a well-motivated identification rather than a completed derivation from classical mechanics. To justify the name physically, we must introduce time evolution and show that localized wave packets move in the classical limit with velocity related to their k-content.

Question from the audience. Does large k mean a large or a small wavelength? [a]

Response. A large $|k|$ gives rapid variation and a short wavelength; a small $|k|$ gives slow variation and a long wavelength, because $L = 2\pi/|k|$.

[a] Lecture timestamp: 01:54:03–01:54:30.

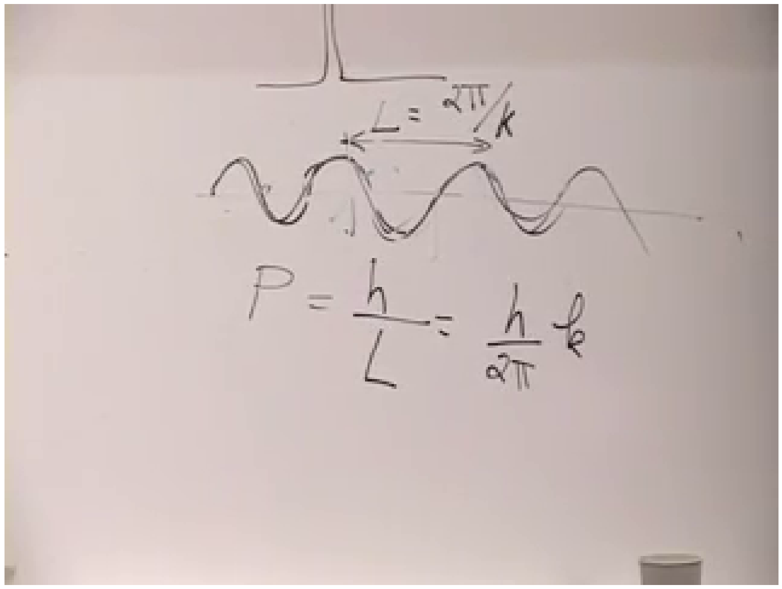

Figure 1.11: Wavelength and the de Broglie identification $p = \hbar k$.

1.8 Position, Momentum, and Incompatible Bases

The position and momentum eigenfunctions could hardly look more different. A position eigenfunction is concentrated at one point and contains no definite k. A momentum eigenfunction extends over the entire line and contains no definite position. The two families form different generalized bases in the same state space.

This is the first glimpse of the uncertainty principle in the language of state vectors. A sharply localized wave packet requires a superposition of many values of k, while a single value of k produces complete delocalization. The precise inequality will come later; the clash of the two eigenbases is already visible.

There is also a useful geometric way to picture the complex plane wave. At each point x, e^{ikx} is a point on the unit circle in the complex plane. As x increases, that point winds around the circle. If the spatial axis is drawn perpendicular to the complex plane, the resulting curve is a helix. Large $|k|$ winds rapidly; small $|k|$ winds slowly.

Question from the audience. Can e^{ikx} be visualized as a function twisting around the x-axis?[a]

Response. Yes. The real and imaginary parts are the two perpendicular coordinates of a unit vector in the complex plane. Advancing along the spatial axis rotates that vector through phase kx, producing the helix described above.

[a]Lecture timestamp: 01:50:04–01:53:05.

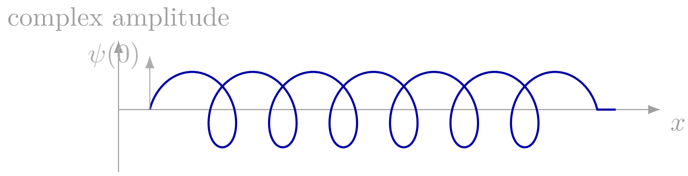

Figure 1.12: Transcript-based schematic of the plane-wave phase winding as a helix along the spatial axis.

Question from the audience. Why was wavefunction collapse absent from the five postulates, and what is the probability distribution for momentum? [a]

Response. The lecture has so far stated only which outcomes are possible and how their probabilities are assigned. The physical measurement process and its effect on the state have not yet been introduced. Likewise, the momentum-space probability rule requires expanding the state in the $|k\rangle$ basis; that construction is deferred to the next step of the course.

[a] Lecture timestamp: 01:53:15–01:54:02.

Question from the audience. Classical momentum involves motion in time. Why has a spatial derivative appeared before any time derivative? [a]

Response. The identification must ultimately be checked through time evolution. A normalizable wave packet forms a localized lump. In the appropriate heavy-particle or classical limit, that lump should move according to the classical equations, with its velocity related to the wave-number content of the packet. Time has not yet entered the theory, so that dynamical check remains ahead.

[a] Lecture timestamp: 01:54:31–01:56:03.